CO2 Footprint Tracker(Architecture)

System Components

- **User Profile Manager**:
    - Responsible for setting up lifestyle attributes and storing user preferences.
- **Activity Logger**:
    - Manages the logging of daily activities and categorizes them by type (e.g., transport, diet).
- **CO_2 Analyzer**:
    - Processes activity logs to calculate CO_2 emissions for each category and aggregates data over different time periods.
- **Optimization Engine**:
    - Generates personalized suggestions based on patterns detected in the user's activity data.

Technology Stack

- **Programming Language**:
    - Java, utilizing OpenJDK 22.
- **GUI Framework**:
    - JavaFX for developing the user interface.
- **Data Storage**:
    - Local storage using MSSQL or JSON files as an embedded database.
- **Rationale**:
    - These choices were made for their simplicity and compatibility with the lab environment, with the option to scale to a more robust database later if necessary.

- **Future Plans**:
    - The prototype will be refined as the project progresses.

Project Management

Project Planning

- **Sprints Overview**:
    - The project is divided into four main sprints:
        - **Sprint 1**: Focus on UI design for lifestyle configuration and activity logging.
        - **Sprint 2**: Develop CO_2 calculation logic and implement data storage.
        - **Sprint 3**: Build analysis features, including data visualization and filtering.
        - **Sprint 4**: Finalize optimization suggestions, conduct testing, and prepare for deployment.
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Tools

- **Version Control and Project Management**:
    - GitLab is used for version control, document management, and issue tracking to ensure effective collaboration and project management.
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Team Organization

- **Defined Roles**:
    - Each team member has a specific role:
        - **Development Lead**: Oversees backend development, ensuring accurate CO_2 calculations and data management.
        - **UI/UX Designer**: Focuses on creating an intuitive user interface.
        - **Data Analyst**: Ensures accuracy in CO_2 factors and suggests optimization algorithms.
        - **Project Manager**: Handles sprint planning, coordinates tasks, and ensures timely progress.